3 OUTCOME-BASED ASSESSMENT (120)

Total Marks 120.00

3.1 Evaluation of Continuous Assessment: Assignments, Unit Tests, Mid-Term, etc. (10)

Total Marks 10.00

Institute Marks : 10.00

The institution adopts a planned and outcome-oriented continuous assessment framework to monitor and evaluate students' learning progress throughout the semester. The evaluation system is designed to ensure fairness, transparency, consistency, and alignment with Course Outcomes (COs), which are further mapped to Program Outcomes (POs) and Program Specific Outcomes (PSOs).

Continuous assessment is conducted as per the approved academic calendar and includes multiple evaluation components to assess students' knowledge, analytical ability, application skills, and professional competencies.

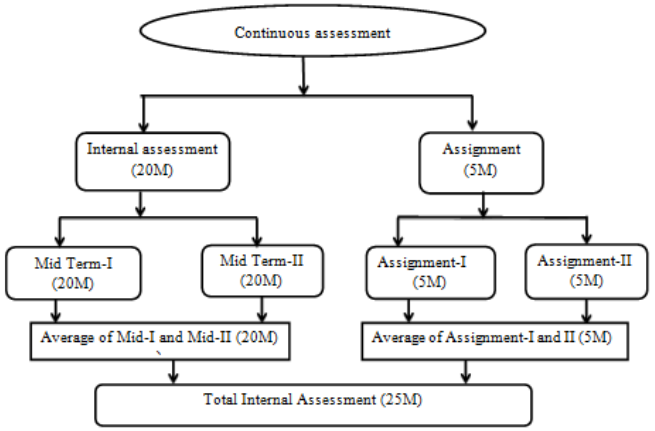
Modes of Continuous Assessment

R18 Regulation:

There are two midterm examinations conducted during the semester. The first midterm examination covers the first 2.5 units of the syllabus. The second midterm examination covers the remaining 2.5 units of the syllabus. This ensures equal distribution of the syllabus across both midterms.

Each midterm examination consists of three assessment components. Part-A includes five short answer questions, with one mark for each question, totalling five marks. Part-B includes three long answer questions with internal choice, with five marks for each question, totalling fifteen marks. In addition, an assignment carrying five marks is included for each midterm examination.

The total marks for each midterm examination are 25. The assignment must be completed before the midterm examination as per the schedule. The marks obtained in both midterm examinations are later used to calculate the final CIE marks by taking their average.



Category	Syllabus Coverage	Assessment Components	Marks
Midterm Examination – I	First 50% of syllabus	Part-A: 5 short answer questions × 1 mark = 5 marks Part-B: 3 long answer questions with internal choice × 5 marks = 15 Marks	20
Midterm Examination – II	Remaining 50% of syllabus	Part-A: 5 short answer questions × 1 mark = 5 marks Part-B: 3 long answer questions with internal choice × 5 marks = 15 marks	20
Assignments	As per syllabus portion taught	Two assignments, each for 5 marks. The average of both is considered.	5
Total CIE	Complete syllabus	Midterm average + Assignment average	**25**

R22 Regulation:

The total marks allotted for CIE are 40 marks. The evaluation is done through midterm examinations, assignments, and a subject-based assessment activity.

There are two midterm examinations conducted during the semester. The first midterm examination covers the first 50 percent of the syllabus. The second midterm examination covers the remaining 50 percent of the syllabus. Each midterm examination is evaluated for 30 marks.

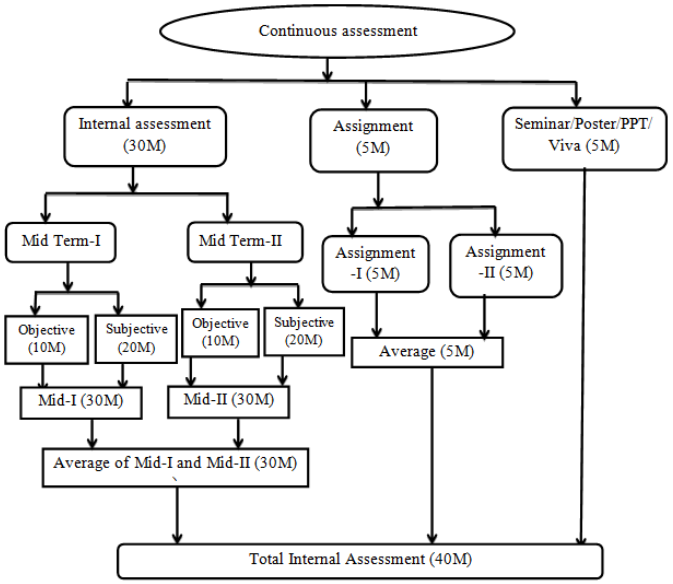
Each midterm examination has two parts. Part-A is an objective or quiz-based test for 10 marks. It includes multiple choice questions, fill-in-the-blanks, and match-the-following questions. Part-B is a descriptive examination for 20 marks. It contains six questions, out of which the student must answer any four questions. Each question carries five marks.

The average of the marks obtained in the two midterm examinations is taken as the final midterm marks for 30 under CIE.

Apart from the midterm examinations, assignments carry five marks. Two assignments are given during the semester. Each assignment is evaluated for five marks. The average of the two assignment marks is taken as the final assignment marks. The first assignment is submitted before the first midterm examination, and the second assignment is submitted before the second midterm examination.

The remaining five marks are allotted for Subject Viva-Voce, PPT, Poster Presentation, or Case Study. This activity is conducted on selected topics from the subject and is completed before the second midterm examination.

Overall, the CIE marks are calculated by combining the average midterm marks, assignment marks, and viva or presentation marks, making a total of 40 marks.



Category	Syllabus Coverage	Assessment Components	Marks
Midterm Examination – I	First 50% of syllabus	Part-A: Objective/Quiz (MCQ, fill-in-the-blanks) – 10 marks Part-B: Descriptive (6 questions, answer any 4 × 5 marks) – 20 marks	30
Midterm Examination – II	Remaining 50% of syllabus	Part-A: Objective/Quiz	30

Category	Syllabus Coverage	Assessment Components	Marks
		(MCQ, fill-in-the-blanks) – 10 marks Part-B: Descriptive (6 questions, answer any 4 × 5 marks) – 20 marks	
Assignments	As per syllabus portion taught	Two assignments, each for 5 marks. Average of both is considered.	5
Subject Viva-Voce / PPT / Poster / Case Study	Entire syllabus / selected topics	One subject-based assessment activity	5
Total CIE	Complete syllabus	Midterm average + assignment average + viva/presentation/case study	**40**

Feedback is collected from students before the mid-I examination and after the completion of the course to evaluate the teaching–learning process. This helps in identifying strengths and areas for improvement in course delivery. The feedback is analyzed systematically and necessary corrective actions are taken. It ensures continuous quality improvement as required. Thus, student feedback plays an important role in enhancing academic effectiveness and achieving desired learning outcomes.

Constructive Alignment of Assessment with COs, POs, and PSOs

- Each course has well-defined Course Outcomes approved by the Department Academic Committee.
- Assessment questions are carefully framed to address specific COs.
- Every question is mapped to:
  - Relevant CO
  - Appropriate Bloom's Taxonomy level
- The distribution of questions ensures

  a. Balanced coverage of lower and higher cognitive levels

  b.Evaluation of both theoretical understanding and practical application

- The CO attainment contributes directly to the PO and PSO attainment, ensuring outcome-based education compliance.

ANURAG ENGINEERING COLLEGE
An Autonomous Institution
(Affiliated to JNTUH-Hyderabad,Approved by AICTE-New Delhi)
ANANTHAGIRI (V) (M), SURYAPET (D), TELANGANA-508206
2024-25

CSE

Assignment-I Question Paper

Class : IV B.Tech I Semester CSE
Course: Linux Programming (R18)
Answer all the questions. 5*5M=25 M

S.No.	Question	COs	BT Level
1	Discuss awk scripts with suitable examples.	CO1	L2
2	Demonstrate the use of file handling utilities by writing a program that performs operations such as creating, reading, writing, and closing a file with a suitable example.	CO1	L3
3	Describe about the file ownership commands available in LINUX.	CO2	L2
4	Apply the appropriate file handling commands by writing a program that demonstrates the syntax and usage of mkdir, rmdir, chdir, opendir, and readdir with suitable examples.	CO2	L3
5	Draw the Memory Layout of C program in Main memory and Explain briefly.	CO3	L2

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2024-25

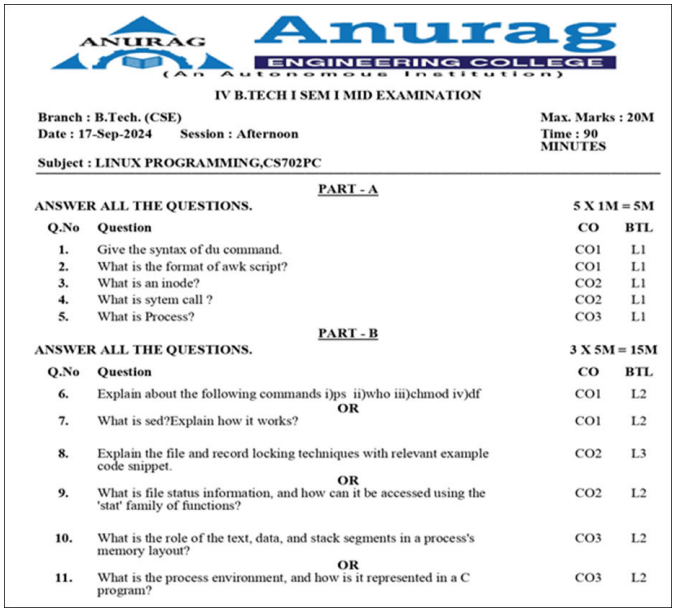
CSE

Assignment-II Question Paper

Class: IV B.Tech I Semester CSE
Course: Linux Programming(R18)
Answer all the questions. 5*5M=25 M

S.No.	Question	COs	BT Level
1	Illustrate the following system calls with examples: a. fork b. vfork c. exit d. exec	CO3	L2
2	Apply interprocess communication techniques by implementing programs that demonstrate the use of Pipes and FIFOs with suitable examples.	CO4	L3
3	Explain IPC using Message Queues with Suitable Program.	CO4	L2
4	Write the uses of Shared memory, system structure, system calls and illustrate with an example.	CO5	L2
5	Implement a C program that uses shared memory for interprocess communication. The program should perform the following tasks: i. Creates a shared memory segment. ii. Attaches to the shared memory segment. iii. Writes some content into the shared memory segment. iv. Reads and prints the content from the shared memory segment.	CO5	L3

Figure 1.3.1.A: Sample Assignments



Anurag ENGINEERING COLLEGE
(An Autonomous Institution)

IV B.TECH I SEM I MID EXAMINATION

Branch : B.Tech. (CSE) — **Max. Marks : 20M**
Date : 17-Sep-2024 **Session : Afternoon** — **Time : 90 MINUTES**

Subject : LINUX PROGRAMMING,CS702PC

PART - A

ANSWER ALL THE QUESTIONS. — **5 X 1M = 5M**

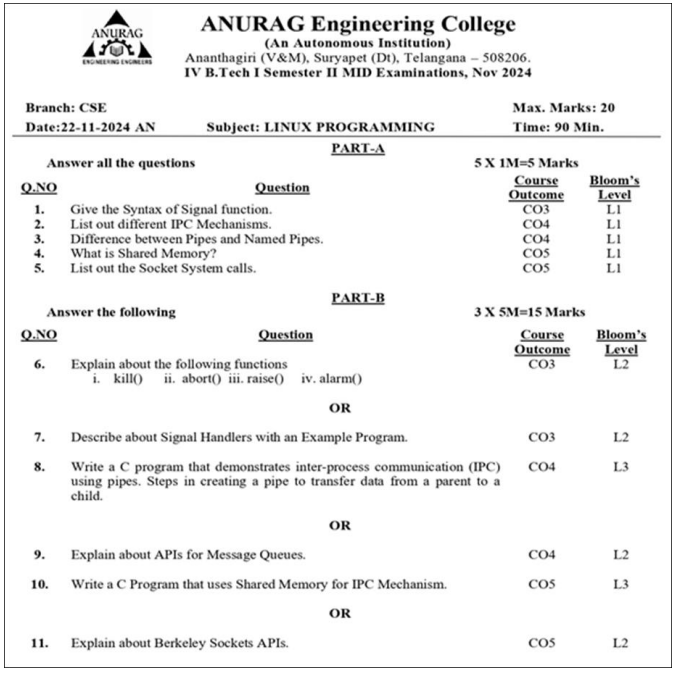
Q.No	Question	CO	BTL
1.	Give the syntax of du command.	CO1	L1
2.	What is the format of awk script?	CO1	L1
3.	What is an inode?	CO2	L1
4.	What is sytem call ?	CO2	L1
5.	What is Process?	CO3	L1

PART - B

ANSWER ALL THE QUESTIONS. — **3 X 5M = 15M**

Q.No	Question	CO	BTL
6.	Explain about the following commands i)ps ii)who iii)chmod iv)df	CO1	L2
	OR		
7.	What is sed?Explain how it works?	CO1	L2
8.	Explain the file and record locking techniques with relevant example code snippet.	CO2	L3
	OR		
9.	What is file status information, and how can it be accessed using the 'stat' family of functions?	CO2	L2
10.	What is the role of the text, data, and stack segments in a process's memory layout?	CO3	L2
	OR		
11.	What is the process environment, and how is it represented in a C program?	CO3	L2

Figure 1.3.1.B: Sample I-Mid Question Paper



ANURAG Engineering College
(An Autonomous Institution)
Ananthagiri (V.&M), Suryapet (Dt), Telangana – 508206.
IV B.Tech I Semester II MID Examinations, Nov 2024

Branch: CSE — **Max. Marks: 20**
Date:22-11-2024 AN **Subject: LINUX PROGRAMMING** — **Time: 90 Min.**

PART-A

Answer all the questions — **5 X 1M=5 Marks**

Q.NO	Question	Course Outcome	Bloom's Level
1.	Give the Syntax of Signal function.	CO3	L1
2.	List out different IPC Mechanisms.	CO4	L1
3.	Difference between Pipes and Named Pipes.	CO4	L1
4.	What is Shared Memory?	CO5	L1
5.	List out the Socket System calls.	CO5	L1

PART-B

Answer the following — **3 X 5M=15 Marks**

Q.NO	Question	Course Outcome	Bloom's Level
6.	Explain about the following functions i. kill() ii. abort() iii. raise() iv. alarm()	CO3	L2
	OR		
7.	Describe about Signal Handlers with an Example Program.	CO3	L2
8.	Write a C program that demonstrates inter-process communication (IPC) using pipes. Steps in creating a pipe to transfer data from a parent to a child.	CO4	L3
	OR		
9.	Explain about APIs for Message Queues.	CO4	L2
10.	Write a C Program that uses Shared Memory for IPC Mechanism.	CO5	L3
	OR		
11.	Explain about Berkeley Sockets APIs.	CO5	L2

Figure 1.3.1.C: Sample II-Mid Question Paper

Assessment Quality Assurance Practices

To maintain the quality of continuous assessment, the department follows these measures:

- Pre-assessment review of question papers to ensure relevance, difficulty balance, and syllabus coverage.
- Uniform evaluation guidelines through predefined rubrics and model solutions.
- Consistency in valuation across all sections handled by different faculty members.
- Timely feedback to students after evaluation to help them identify learning gaps.
- Post-assessment analysis of performance to evaluate attainment levels of COs.

Performance Analysis and Improvement Measures

- Marks obtained in continuous assessments are analyzed to determine CO attainment levels.
- Courses with lower attainment are identified.
- Improvement actions such as:
  - Remedial instruction
  - Additional practice sessions
  - Re-teaching of difficult concepts are implemented.
- The effectiveness of these actions is reviewed in subsequent assessments.

3.2 Evaluation of the Semester End Exam (SEE) Question Paper (10)	Total Marks 10.00
	Institute Marks : 10.00

The Semester End Examination (SEE) is used as a summative assessment tool to measure the extent of attainment of Course Outcomes (COs) at the end of the semester. To ensure quality, fairness, and outcome-based evaluation, the institution follows a systematic procedure for the setting, review, and evaluation of SEE question papers, with explicit alignment to Program Outcomes (POs) and Program Specific Outcomes (PSOs).

Process of Setting the SEE Question Paper:

- **Selection of Question Paper Setters:**
  SEE question papers are prepared by external subject experts from reputed institutions as approved by the BoS. This ensures academic competence and impartiality.
- **Question Paper Blueprint:**
  A specification grid (blueprint) is provided to the paper setter indicating:
  - Unit-wise syllabus coverage
  - CO-wise weightage
  - Marks distribution
- **Cognitive Level Distribution:**
  Questions are framed according to the Revised Bloom's Taxonomy, covering multiple cognitive levels such as understanding, application, analysis, evaluation, and design. This ensures assessment of both fundamental knowledge and higher-order thinking skills.
- **Confidentiality and Security:**
  Strict confidentiality is maintained during paper setting and submission. Question papers are submitted through secure channels to the CoE to avoid any possibility of leakage.

For each course 4 sets of Question papers will be collected from different experts from the panel. Out of 4 sets one set will be selected. The selected set will be considered for conducting the exam on that day

The concerned subject Co-ordinator will be called for moderation of the question paper two hours prior to the examination. During this process, the question paper will be reviewed to ensure that all questions are within the prescribed syllabus, the distribution of marks is appropriate, and the questions are properly aligned with Bloom's Taxonomy levels.

Quality Assurance in SEE Question Paper Setting

To maintain academic rigor and uniform standards, the following quality measures are adopted:

1.Question Paper Blueprint

- A blueprint is prepared specifying CO-wise wieght, Blooms taxonomy distribution and Unit-wise coverage. This ensures balanced and unbaised assessment

2.Moderation

The drafted question paper is reviewed by senior faculty for:

- CO and Bloom's level mapping
- Difficulty level balance
- Clarity of questions
- Syllabus coverage
- Necessary corrections are incorporated before final submission.

3.Confidentiality and Standardization

- The question paper preparation and submission follow strict confidentiality norms.

Evaluation Process of SEE Answer Scripts

- Model answers and marking schemes are prepared prior to evaluation.
- Evaluation is carried out by qualified and experienced examiners.
- Uniform valuation practices are ensured by:
  - Clear rubrics
  - Pre-valuation meetings (where applicable)
- Random verification and re-checking are carried out to minimize subjectivity.

Constructive Alignment with COs, POs, and PSOs

- Each course has clearly articulated Course Outcomes mapped to POs and PSOs.
- While framing the SEE question paper, faculty ensure that:
  - All COs are adequately represented.
  - Questions address different cognitive levels as per Bloom's Taxonomy.
- Every question in the SEE paper is mapped to:
  - Corresponding CO
  - Bloom's level
  - Marks allocation
- This structured mapping ensures that the SEE evaluates knowledge, analytical ability, design skills, and problem-solving competence, contributing to overall program outcome attainment.

R18 Regulation Question Paper Blueprint

Part	No. of Questions	Marks per Question	Total Marks	Remarks
Part – A	10 Questions	2 Marks (5 Questions)3 Marks (5 Questions)	25 Marks	Compulsory. Two questions from each Unit (1 question for 2 marks and 1 question for 3 marks per unit).
Part – B	5 Questions (with internal choice)	10 Marks each	50 Marks	One question from each Unit. Each question has an **"Either/Or"** choice (2 questions per unit; student answers any one).

Unit	CO Coverage	Part – A (Compulsory)	Marks (A)	Part – B (Either / Or Choice)	Marks (B)	Total per Unit
Unit – I	CO – 1	1 × 2M + 1 × 3M	5 Marks	1 × 10M (Answer any one of two questions)	10 Marks	15 Marks
Unit – II	CO – 2	1 × 2M + 1 × 3M	5 Marks	1 × 10M (Answer any one of two questions)	10 Marks	15 Marks
Unit – III	CO – 3	1 × 2M + 1 × 3M	5 Marks	1 × 10M (Answer any one of two questions)	10 Marks	15 Marks
Unit – IV	CO – 4	1 × 2M + 1 × 3M	5 Marks	1 × 10M (Answer any one of two questions)	10 Marks	15 Marks
Unit – V	CO – 5	1 × 2M + 1 × 3M	5 Marks	1 × 10M (Answer any one of two questions)	10 Marks	15 Marks
Total	CO1–CO5	5 Units × (2M + 3M)	**25 Marks**	5 Units × 10M	**50 Marks**	**75 Marks**

R22 Regulation Question Paper Blueprint

Part	No. of Questions	Marks per Question	Total Marks	Remarks
Part – A	10 Questions	1 Mark (10 Questions)	10 Marks	Compulsory. Two questions from each Unit (1 Mark for each question).
Part – B	5 Questions (with internal choice)	10 Marks each	50 Marks	One question from each Unit. Each question has an **"Either/Or"** choice (2 questions per unit; student answers any one).

Unit	CO Coverage	Part – A (Compulsory)	Marks (A)	Part – B (Either / Or Choice)	Marks (B)	Total per Unit
Unit – I	CO – 1	2 × 1M Questions	2 Marks	1 × 10M (Answer any one of two questions)	10 Marks	12 Marks

Unit	CO Coverage	Part – A (Compulsory)	Marks (A)	Part – B (Either / Or Choice)	Marks (B)	Total per Unit
Unit – II	CO – 2	2 × 1M Questions	2 Marks	1 × 10M (Answer any one of two questions)	10 Marks	12 Marks
Unit – III	CO – 3	2 × 1M Questions	2 Marks	1 × 10M (Answer any one of two questions)	10 Marks	12 Marks
Unit – IV	CO – 4	2 × 1M Questions	2 Marks	1 × 10M (Answer any one of two questions)	10 Marks	12 Marks
Unit – V	CO – 5	2 × 1M Questions	2 Marks	1 × 10M (Answer any one of two questions)	10 Marks	12 Marks
Total	CO1–CO5	10 × 1M Questions	**10 Marks**	5 × 10M Questions	**50 Marks**	**60 Marks**

Sample Question Paper:

Question Paper Code: R18A41CS01

R18

ANURAG Engineering College

(An Autonomous Institution)

IV B. Tech I Semester Regular/Supplementary Examinations, Dec-2024

LINUX PROGRAMMING

(COMPUTER SCIENCE ENGINEERING)

Time: 3 Hours **Max. Marks: 75**

Section – A (Short Answer type questions) (25 Marks)

Answer All Questions

		Course Outcome	B.T Level	Marks
1.	What is the importance of dot/(.) and dot dot/(..) directories in the file system?	CO1	L1	2M
2.	Explain about I/O redirection operations in shell?	CO1	L2	3M
3.	How fcntl() system call is used in file system?	CO2	L1	2M
4.	Compare file locking with record locking?	CO2	L2	3M
5.	List any four signal functions?	CO3	L1	2M
6.	Explain how to terminate process normally or abnormally?	CO3	L2	3M
7.	Write down the advantages and disadvantages of FIFOs?	CO4	L1	2M
8.	Extend with an example describe about pipe () system call?	CO4	L2	3M
9.	How stream sockets are different form data sockets?	CO5	L1	2M
10.	Summarize the race conditions with shared memory?	CO5	L2	3M

Section B (Essay Questions)

Answer all questions, each question carries equal marks. **(5 X 10M = 50M)**

11. A)	With the help of syntax and example explain the various process utility commands.	CO1	L3	10M
	OR			
B)	Develop a shell script to count the specified number of lines in a text file without using wc command?	CO1	L3	10M
12. A)	Make use of system calls stat( ), fstat( ) and lstat( ) in a shell script with any one example program.	CO2	L3	10M
	OR			
B)	Examine and elaborate various functions used in listing the content of a directory with syntax and sample code?	CO2	L3	10M
13. A)	Distinguish between fork() and exec() system calls with example code?	CO3	L3	10M
	OR			
B)	Inspect the mechanism for handling a signal with example code?	CO3	L3	10M
14. A)	Construct a program and explain how to transfer a large amount of data between two processes using message queues?	CO4	L3	10M
	OR			
B)	Dissect how process synchronization is achieved using semaphores with examples?	CO4	L3	10M
15. A)	Identify the system functions associated for creating and destroying a shared memory with examples?	CO5	L3	10M
	OR			
B)	Distinguish the socket system calls for connection oriented and connectionless protocol?	CO5	L3	10M

Question Paper Code: CS503PC

R22

ANURAG Engineering College

(An Autonomous Institution)

III B.Tech I Semester Regular/Supplementary Examinations, Nov/Dec - 2025

DEVOPS

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 Hours **Max. Marks: 60**

Section – A (Short Answer type questions) **(10 Marks)**

Answer All Questions	Course Outcome	B.T Level	Marks
1. What is DevOps?	CO1	L1	1M
2. Explain the Agile development model?	CO1	L2	1M
3. What is the DevOps lifecycle for business agility?	CO2	L1	1M
4. What is continuous testing in DevOps?	CO2	L1	1M
5. What is the need for source code control in project management?	CO3	L1	1M
6. What are hosted Git servers?	CO3	L1	1M
7. What is the role of a build system in software development?	CO4	L1	1M
8. What is Jenkins, and why is it used?	CO4	L1	1M
9. What are the pros of automated testing?	CO5	L1	1M
10. What is Selenium used for?	CO5	L1	1M

Section B (Essay Questions)

Answer all questions, each question carries equal marks. **(5 X 10M = 50M)**

11. Explain the Agile Development Model and its significance in modern software development.	CO1	L2	10M
OR			
12. Compare and contrast Scrum and Kanban methodologies in Agile development.	CO1	L2	10M
13. Explain the DevOps Lifecycle and its role in achieving Business Agility.	CO2	L2	10M
OR			
14. Analyze why microservices are more suitable for DevOps practices and modern software development.	CO2	L3	10M
15. Explain the need for source code control in modern software development.	CO3	L2	10M
OR			
16. Explain how Docker integrates with source code management tools to streamline deployment.	CO3	L2	10M
17. Explain the concept of build systems and their importance in software development.	CO4	L2	10M
OR			
18. Compare alternative build servers to Jenkins and their approach to infrastructure as code	CO4	L2	10M
19. Explain the various types of testing and their importance in the software development life cycle (SDLC).	CO5	L2	10M
OR			
20. Explain test-driven development (TDD) and REPL-driven development as modern software development practices	CO5	L2	10M

3.3 Evaluation of Laboratory Work and Workshop (Continuous and SEE) (10) **Total Marks 10.00**

Institute Marks : 10.00

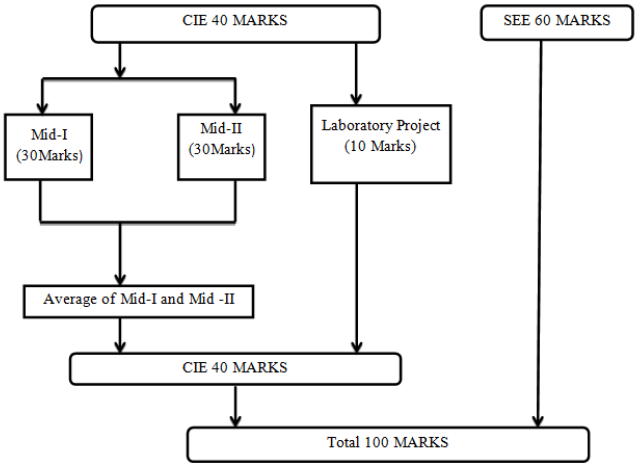
The evaluation of laboratory and workshop courses is carried out through a well-structured Continuous Internal Evaluation (CIE) and Semester End Examination (SEE) to ensure effective assessment of students' practical skills, conceptual understanding, analytical ability, and professional competencies. Continuous assessment focuses on students' regular performance throughout the semester, including attendance, active participation, and timely completion of experiments, maintenance of laboratory records, observation skills. It also evaluates the accuracy of results, understanding of concepts, and ability to handle tools, This on-going evaluation encourages consistency, discipline, and continuous improvement in practical learning.

The Semester End Examination (SEE), on the other hand, assesses the overall practical competency of students at the end of the course. It typically includes the performance of assigned experiments , viva voce (oral examination), submission of practical records, and demonstration of technical skills within a specified time. The SEE evaluates students' ability to apply theoretical knowledge in practical situations, interpret results, troubleshoot errors. Together, Continuous Assessment and SEE provide a balanced and fair evaluation system that measures both process and performance, ensuring that students develop the required practical proficiency and professional competence

The evaluation process is aligned with the Course Outcomes (COs) and mapped to relevant Program Outcomes (POs) and Program Specific Outcomes (PSOs).

R22 Regulation:

The evaluation of Laboratory Work is carried out through both Continuous Assessment and the Semester End Examination (SEE) .The total marks are generally divided into Continuous Assessment (40%) and Semester End Examination (60%).



1. Continuous Internal Evaluation (CIE):

Continuous assessment is conducted throughout the semester to monitor and enhance student learning.

1. A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks

2. 10 marks for viva-voce (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.

3. Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 10 marks. The following components and rubrics are used:

Components of CIE

Component	Marks
Laboratory/Workshop Write-Up	10
Program Execution / Practical Performance	10
Viva-Voce	10
Total (CIE)	**30**

Rubrics for Continuous Internal Evaluation -30 Marks

Component	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
1. Lab Record / Write-Up	Problem statement clearly defined; algorithm/flowchart well structured; code documented with comments; test cases included; outputs verified; result justified	Minor gaps in algorithm or documentation; limited test cases; minor formatting issues	Problem partially defined; incomplete algorithm; minimal comments; limited testing	Poor documentation; missing algorithm; no test cases; incorrect or incomplete outputs
2. Program Development & Execution	Program compiles/runs without errors; correct logic; proper use of data structures/functions; handles edge cases; efficient implementation	Minor syntax/runtime errors corrected independently; logic mostly correct; limited edge case handling	Requires assistance to debug; partially correct logic; inefficient implementation	Unable to execute; major logical errors; incorrect output
3. Viva-Voce / Conceptual Understanding	Strong understanding of concepts (syntax, logic, complexity); explains code clearly; answers application-based questions confidently	Good understanding; answers most questions correctly; minor conceptual gaps	Basic understanding; difficulty explaining logic; answers direct questions only	Unable to explain code or concepts; lacks clarity

Laboratory Project:

The remaining 10 marks are for Laboratory Project, which consists of the Design (or) Software / Hardware Model Presentation (or) App Development (or) Prototype Presentation submission which shall be evaluated after completion of laboratory course and before semester end practical examination

Rubrics for Laboratory Project – 10 Marks

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
Problem Identification & Objectives	Clear problem statement; well-defined objectives; strong relevance	Problem defined with minor gaps	Problem partially defined; objectives unclear	Problem not clearly defined
Design / Model Development	Logical, innovative, and technically sound design/model	Design mostly correct with minor issues	Basic design; lacks clarity/optimization	Incomplete or incorrect design/model
Implementation / Functionality	Fully functional with accurate results/output	Functional with minor errors	Partially functional	Not working / major errors
Innovation & Creativity	Highly innovative; demonstrates original thinking	Some innovative elements	Limited creativity	No innovation shown
Presentation & Demonstration	Clear explanation, confident presentation, answers questions effectively	Good explanation with minor hesitation	Basic explanation; limited clarity	Unable to explain/demonstrate properly
Documentation / Report	Well-structured report with diagrams, results, and conclusion	Minor omissions in documentation	Incomplete documentation	No proper documentation

2. Semester End Examination (SEE) :

Components of SEE

Component	Marks
Write-Up	10
Experiment / Program Evaluation	15
Program Execution	15
Program Presentation	10
Viva-Voce	10
Total	**60**

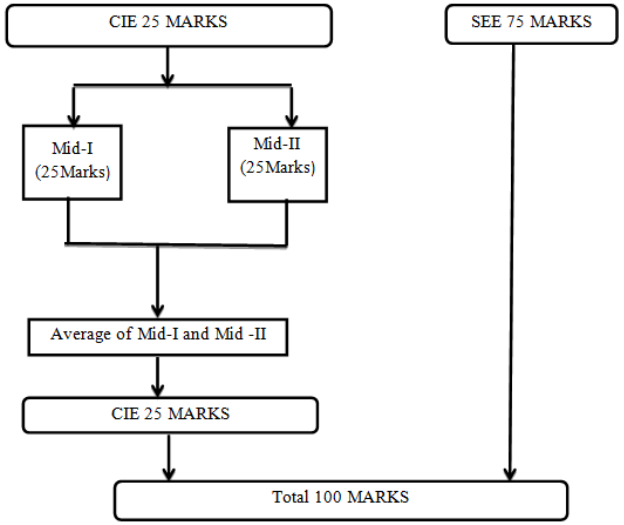
Rubrics for Semester End Examination

Component	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
1. Write-Up	Clear problem definition; well-structured algorithm/design; proper flowchart/ER/UML (where applicable); relevant test cases; correct output; neat presentation	Minor omissions in design/test cases; small formatting errors	Problem partially defined; incomplete algorithm/design; limited testing	Incorrect or missing documentation; no proper design
2. Experiment / Program Evaluation	Correct logic; appropriate data structures/techniques used; optimized solution; handles edge cases; strong analytical approach	Mostly correct logic; minor inefficiencies; limited edge case handling	Partially correct logic; requires guidance; inefficient implementation	Incorrect approach; major conceptual errors
3. Program Execution	Program compiles and runs without errors; accurate output; independent debugging; proper input handling	Minor syntax/runtime errors corrected; output mostly correct	Executes partially; requires assistance; incomplete output	Unable to execute; incorrect output
4. Program Presentation	Code well-structured and modular; meaningful variable names; proper indentation; clear explanation of logic; confident presentation	Mostly structured code; minor formatting issues; explanation adequate	Poor structure; limited clarity in explanation	Unorganized code; unable to explain program
5. Viva-Voce	Thorough conceptual clarity; explains complexity, tools, and implementation choices; answers application-based questions confidently	Good understanding; minor conceptual gaps	Basic understanding; difficulty explaining advanced aspects	Unable to answer fundamental questions

R18 Regulation:

The students record the output in their observation after implementing the relevant program. The Course Instructor evaluates the students' performance every week. The distribution of laboratory shall be 25 marks for internal evaluation (15 marks for day-to-day work and 10 marks for internal tests) and 75 marks for end examination. There shall be two midterm examinations in a semester and the average of the two shall be considered for the award of marks for midterm examinations. The 15 marks for day-to-day work shall be considered only if the student attends the

midterm examination when conducted. An Internal lab exam of 3 hours is conducted to assess the ability of a student performance. The end semester practical examination is conducted for 3 hour duration.



1. Continuous Internal Evaluation (CIE):

Continuous assessment is conducted throughout the semester to monitor and enhance student learning. The following components and rubrics are used:

Components of CIE

Component	Marks
Day-to-Day Work	15
Program Execution / Practical Performance in CIE	10
Total (CIE)	**25**

Rubrics for Continuous Internal Evaluation:

A. Day-to-Day Work – 15 Marks

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
Preparation & Understanding	Clearly understands problem; explains logic confidently	Minor gaps in understanding	Basic idea present but lacks clarity	Unable to explain logic

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
Program Implementation	Correct, efficient, and well-structured code	Minor syntax/logic errors	Multiple errors; needs guidance	Incomplete / incorrect program
Observation Record	Neat, complete outputs with proper format	Minor omissions	Incomplete outputs / untidy record	Not submitted / largely incomplete
Regularity & Participation	Consistent attendance and active participation	Regular but less interactive	Irregular participation	Poor attendance / no participation
Timely Submission	Always on time	Minor delays	Frequent delays	Not submitted

B. Program Execution / Practical Performance in CIE – 10 Marks

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)
Problem Analysis	Correctly understands and plans solution	Minor clarification required	Partial understanding	Misinterprets problem
Coding Accuracy	Error-free execution with correct output	Minor errors corrected independently	Errors corrected with assistance	Unable to execute correctly
Debugging Skills	Identifies and fixes errors independently	Fixes with minimal help	Requires guidance	Unable to debug
Output & Result Verification	Accurate output with proper validation	Minor mistakes in output	Output partially correct	Incorrect / no output

2. Semester End Examination (SEE) :

Components of SEE

Component	Marks
Observation & Record	10
Write-Up	15
Execution	20

Component	Marks
Output	15
Viva-Voce	15
Total	**75**

Rubrics for Semester End Examination

Component	Excellent (4)	Good (3)	Satisfactory (2)	Poor(1)
Observation & Record	Record complete with aim, algorithm, program, output and observations neatly written.	Minor missing details but mostly complete.	Record partially completed with limited details.	Record incomplete or not maintained properly.
Write-Up	Clear problem statement, algorithm/flowchart, program logic and test cases documented neatly.	Minor omissions in explanation or formatting.	Basic explanation with limited clarity.	Incomplete or unclear documentation.
Execution	Program compiles and runs correctly with proper logic and no errors.	Minor syntax or logical errors but program executes.	Program runs partially with incorrect logic.	Program fails to compile or execute.
Output	Correct output obtained for all test cases and verified properly.	Output mostly correct with minor mistakes	Output partially correct.	Output incorrect or not produced.
Viva-Voce	Answers all questions clearly showing strong conceptual understanding.	Answers most questions correctly.	Basic understanding with partial answers.	Unable to answer questions.

3.4 Evaluation of Industrial Training/ Internship (Continuous and SEE) (10) — **Total Marks 10.00**

Institute Marks : 10.00

Our curriculum has Industrial training (or) Internship (or) Industry oriented Mini-project in collaboration with an industry of their specialization.Students can gain practical exposure to real-world industry practices through structured Industry Training and Internship programs**.** These opportunities help students to understand professional work environments, apply theoretical knowledge to practical problems, develop technical and soft skills, and enhance their employability

Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break and during III Year without effecting regular course work. Internship at reputed organization (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester

examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be NO internal marks for Industry Training (or) Internship (or) Industry Oriented Mini Project.

Evaluation Process:

The evaluation process for internship or industrial training is carried out through a structured system of two internal reviews followed by a final externalexpert review to ensure comprehensive assessment. The first review isconducted at the midway stage of the internship, where students present their progress, tasks performed, and preliminary learning outcomes. Feedback is provided by the faculty mentor to guide improvements. The second review isconducted at the end of the internship, where students submit a detailed reportand deliver a presentation or viva voce, demonstrating the work completed andskills acquired. Finally, a review by an external expert from industry is conducted to provide an unbiased evaluation of the student'sperformance, technical knowledge, and practical exposure. The final assessmentis based on including internal reviews, external expert feedback, and supporting documents, ensuring a fair and outcome-based evaluation process.

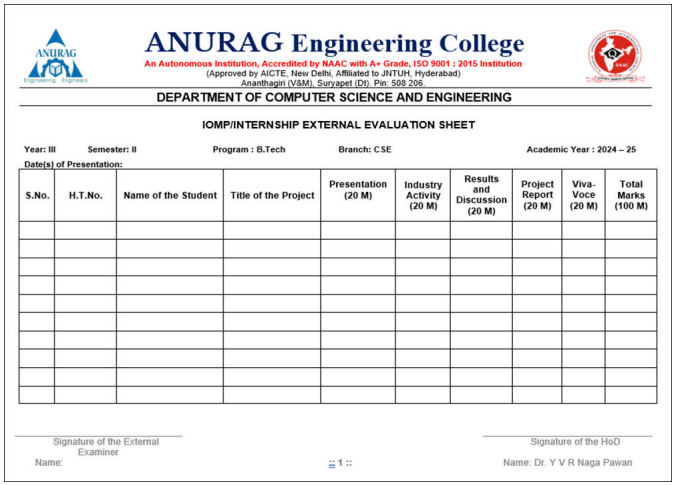
POs and PSOs Addressed:

SNO	Name of the Course	POs Addressed	PSOs Addressed
1	(CS606PW)-Internship	PO6,PO7,PO8,PO9,PO10, PO11	PSO1,PSO2

Rubrics for Industrial training (or) Internship (or) Industry oriented Mini-project

Component	Excellent (4)	Good (3)	Average (2)	Poor (1)
Presentation	Highly effective delivery, well-structured content, excellent visuals, confident communication	Clear delivery with minor issues, good structure and visuals	Basic presentation, some clarity issues, limited engagement	Poor delivery, unclear content, lack of preparation
Industry Activity	Strong industry relevance, active participation, excellent skill application, proper documentation	Good relevance and participation, adequate skills	Limited industry exposure, partial participation	No clear relevance, minimal involvement
Results and Discussion	In-depth analysis, accurate results, strong interpretation linked to objectives	Good analysis with minor gaps	Basic results with limited interpretation	Incorrect or missing analysis
Project Report	Well-organized, detailed, error-free, proper formatting and references	Mostly complete and organized with minor errors	Basic report with some missing elements	Poorly written, incomplete, lacks structure
Viva-Voce	Excellent subject knowledge, confident and accurate responses, strong reasoning	Good understanding, mostly correct answers	Basic knowledge, some difficulty answering	Poor understanding, unable to answer questions

Evaluation for Industrial training (or) Internship (or) Industry oriented Mini-project:



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

IOMP/INTERNSHIP EXTERNAL EVALUATION SHEET

Year: III Semester: II Program : B.Tech Branch: CSE Academic Year : 2024 – 25

Date(s) of Presentation:

S.No.	H.T.No.	Name of the Student	Title of the Project	Presentation (20 M)	Industry Activity (20 M)	Results and Discussion (20 M)	Project Report (20 M)	Viva-Voce (20 M)	Total Marks (100 M)

Signature of the External Examiner
Name:

:: 1 ::

Signature of the HoD
Name: Dr. Y V R Naga Pawan

3.5 Evaluation of Projects (20)

Total Marks 20.00

Institute Marks : 20.00

B.Tech Project Work shall be carried out in two stages: Project Stage – I during IV Year I Semester, Project Stage – II during IV Year II Semester. Each stage will be evaluated for 100 marks. The student has to submit a project work report at the end of each semester. The first report includes project work carried out in IV Year I semester, and the second report includes project work carried out in IV Year I & II Semesters. SEE for both project stages shall be completed before the commencement of SEE Theory examinations. Several reviews are conducted to monitor the progress and evaluate the project.

For Project Stage-I, the departmental committee consists of Head of the Department, project supervisor and a senior faculty member shall evaluate the project work for 75 marks and project supervisor shall evaluate for 25 marks.

Internal Evaluation:

- The department forms Project Review Committee (PRC) every year and it consists of the Head of the department , senior faculty members and project coordinator as members.
- Aproject coordinator is appointed by the Head of the Department who is responsible for planning, scheduling and execution of all activities related to the project.
- The project coordinator instructs the students to select the project domain and submit the synopsis to guide adhering to the timelines decided by the HOD.
- The department encourages the students to carry out in-house projects and required support is provided through continuous hands-on training by internal as well as external experts.
- The students are asked to meet their respective guides regularly and asked to explain the progress in their project.
- Project reviews are conducted regularly by the PRC of the department in the presence of respective guides to check the status of the projects and time to time assessment is done for all the projects.
- Project teams have to submit the project report in the prescribed format given by the institution.

For Project Stage-II, the external examiner shall evaluate the project work for 75 marks and the project supervisor shall evaluate it for 25 marks.For conducting viva-voce of project stage – II, Principal selects an external examiner from the Panel of Experts submitted by Chairman, BoS / HoD of the department.

External Evaluation:

An end semester project, viva voce is conducted with the panel of internal and external examiners. Project reviews are conducted regularly by the PRC of the department in the presence of respective guide to check the status of the projects and time to time assessment is done for all the projects.Project teams have to submit the project report in the prescribed format given by the institution.

Stage-I First review	Problem Statement, Existing system, Proposed system and requirements will be reviewed
Stage-I Second review	Workflow design will be reviewed.
Stage-I Evaluation	The methodology of the project will be reviewed.
Stage-II First review	Modules of the Project and Implementation of project will be reviewed.
Stage-II Second review	Testing, Execution and future scope of the project will be reviewed.
Stage-II Evaluation	Total Project work will be reviewed.

Process to assess individual and team performance

The performance of the individual team member of the project is assessed at the time of presentation in reviews by considering the following criterions:

- Knowledge on the Project work
- Role in the project work
- Communication
- Competency

The performance of the project team is assessed by considering the following criterions:

- Knowledge of the each member and their contribution towards the project
- Coordination in deploying the work
- Schedule management

Rubrics Used for Project Stage-I Evaluation:

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2024-25

CSE

RUBRICS FOR PROJECT STAGE-I (CS705PW):

- CO1: Able to identify a real-world research problem and articulate its scope and significance.
- CO2: Able to propose a feasible solution and differentiate it from existing approaches.
- CO3: Able to conduct a literature survey and establish technical & functional system requirements.
- CO4: Able to design the system architecture and decompose it into functional modules.
- CO5: Able to create design representations and data models for the proposed system.

PROJECT STAGE-I (CS705PW) EVALUATION:

INTERNAL EVALUATION BY SUPERVISOR #REVIEW-I(25 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Problem Understanding**	Clear, precise, well-defined problem and objectives	Minor gaps	Basic understanding	Unclear
2	**Literature Survey**	Multiple relevant sources, critical analysis	Adequate references	Few sources	No proper survey
3	**Feasibility& Specification**	Covers all aspects (technical, economic, operational)	Covers most	Partial	Missing
4	**Presentation**	Clear, confident, structured	Good	Basic	Poor

INTERNAL EVALUATION BY SUPERVISOR #REVIEW-II (25 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Architecture**	Clear, scalable, well-designed	Good structure	Basic	Poor
2	**Modules**	Well-defined with proper functionality	Mostly clear	Limited clarity	Poor
3	**Design Representation**	Proper diagrams (UML/DFD) with accuracy	Minor errors	Basic diagrams	Missing
4	**Presentation**	Clear explanation with demo	Good	Basic	Weak

INTERNAL EVALUATION BY DEPARTMENTAL COMMITTEE (75 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Problem Understanding & Objectives**	Clearly defines a real-world problem with strong justification; objectives are precise, measurable, and well-aligned	Problem is clear; objectives are relevant but lack full clarity/measurability	Problem partially defined; objectives vague or loosely connected	Problem unclear; objectives missing or irrelevant
2	**System Architecture & Design**	Well-structured architecture; clear modular design; detailed UML/DFD; efficient data models	Good design with modules and basic diagrams; minor gaps	Basic design present; lacks modular clarity or detailed diagrams	No clear architecture; incomplete or missing design
3	**Presentation**	Clear, confident delivery; well-organized slides; strong explanation and demo; answers confidently	Good presentation; minor issues in clarity or flow	Basic presentation; limited clarity and explanation	Poor, unorganized; unable to explain project
4	**Project Report**	Comprehensive, well-structured report; all sections included; clear results; error-free	Good report; minor issues in formatting/content	Incomplete or lacks depth; multiple issues	Poorly written; missing key sections
5	**Viva-Voce**	Thorough understanding; answers all questions accurately and confidently	Good understanding; answers most questions	Basic understanding; limited answers	Poor understanding; unable to answer

PROJECT STAGE-I (CS705PW):

CO-PO Mapping:

MAPPING	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11
CO-1						2	1	1	2	-	2
CO-2						2	2	1	2	1	2
CO-3						1	1	1	2	-	2
CO-4						2	1	2	2	3	2
CO-5						1	2	2	2	1	2

CO-PSO Mapping:

MAPPING	PSO-1	PSO-2
CO-1	1	1
CO-2	2	3
CO-3	2	2
CO-4	3	3
CO-5	3	3

Rubrics Used for Project Stage-II Evaluation:

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ANANTHAGIRI (V) (M), SURYAPETA (D), TELANGANA-508206

2024-25

CSE

RUBRICS FOR PROJECT STAGE-II (CS802PW):

- CO1: Able to select appropriate technologies and implement the designed solution through code
- CO2: Able to plan and execute systematic testing using defined methodologies and test cases
- CO3: Able to analyze, interpret, and present results demonstrating solution effectiveness
- CO4: Able to plan, implement, and execute project deployment in a real or simulated environment
- CO5: Able to write an effective technical report and demonstrate the complete project through presentation

PROJECT STAGE-II (CS802PW) EVALUATION:

INTERNAL EVALUATION BY SUPERVISOR #REVIEW-I(25 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Implementation**	Complete implementation with correct logic and innovation	Mostly correct with minor issues	Partial implementation	Incomplete / incorrect
2	**Testing**	Thorough testing with multiple cases and validation	Adequate testing	Limited testing	No proper testing
3	**Presentation**	Very clear, structured, confident explanation	Good clarity	Basic explanation	Poor communication

INTERNAL EVALUATION BY SUPERVISOR #REVIEW-II (25 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Results**	Accurate results with proper analysis and interpretation	Correct results with minor gaps	Partial results	Incorrect / no results
2	**Conclusion &Future Scope**	Conclusion clearly summarizes objectives, methodology, and key results; strong insights; future scope is realistic, innovative, and technically feasible	Conclusion covers major points; future scope is relevant but lacks depth	Basic summary with limited linkage to results; future scope is generic	Weak or missing conclusion; no meaningful future scope
3	**Presentation**	Excellent delivery with strong understanding	Good delivery	Basic explanation	Weak presentation

EXTERNAL EVALUATION BY EXTERNAL EXAMINER (75 Marks)					
	Criteria	**Excellent (4)**	**Good (3)**	**Average (2)**	**Poor (1)**
1	**Presentation**	Clear, well-structured, confident delivery with strong visuals and complete understanding	Good clarity and organization, minor gaps	Basic explanation with limited clarity	Poor communication, unclear or unorganized
2	**Execution**	Fully functional project with correct logic, innovation, and efficient implementation	Mostly functional with minor issues	Partially working project	Incomplete or non-working
3	**Results**	Accurate results with deep analysis, interpretation, and justification	Correct results with some analysis	Limited or basic interpretation	Incorrect or missing results
4	**Project Report**	Comprehensive, well-structured, error-free, with diagrams, references, and proper formatting	Good documentation with minor issues	Basic report with missing details	Poor or incomplete report
5	**Viva-Voce**	Excellent subject knowledge, answers confidently and accurately	Good understanding with minor errors	Basic understanding	Unable to answer questions

PROJECT STAGE-II (CS802PW):

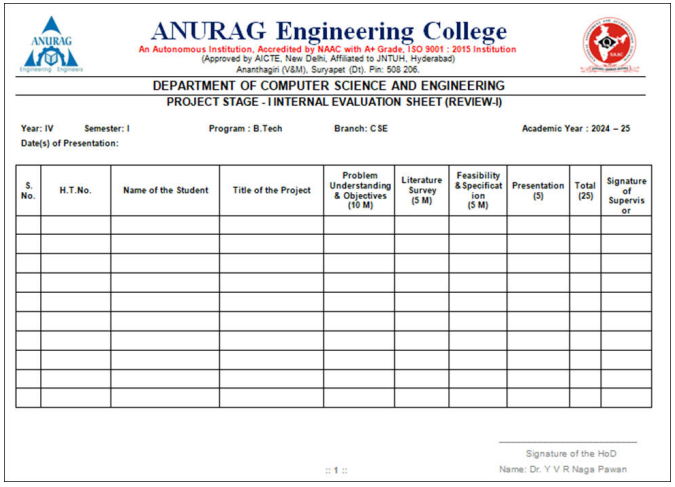
CO-PO Mapping:

MAPPING	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11
CO-1						2	1	1	1	2	3
CO-2						2	2	1	1	2	2
CO-3						1	1	1	3	1	3
CO-4						2	1	2	2	3	2
CO-5						1	2	2	3	2	2

CO-PSO Mapping:

MAPPING	PSO-1	PSO-2
CO-1	3	3
CO-2	3	3
CO-3	2	2
CO-4	3	3
CO-5	3	2

Sample Evaluation Sheets:



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT STAGE - I INTERNAL EVALUATION SHEET (REVIEW-I)

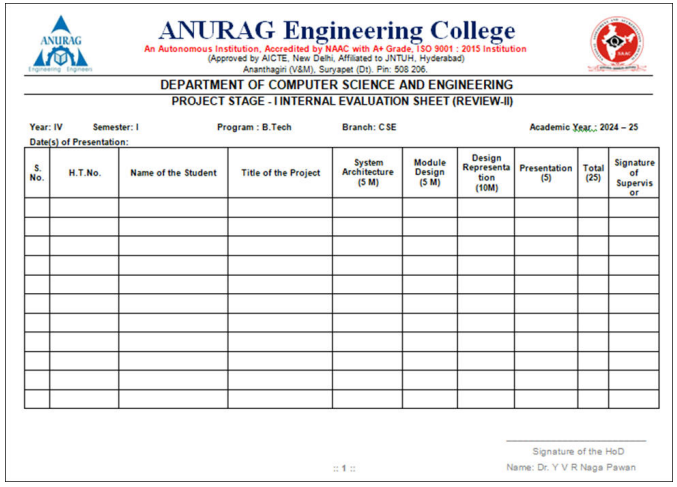
Year: IV Semester: I Program : B.Tech Branch: CSE Academic Year : 2024 – 25

Date(s) of Presentation:

S. No.	H.T.No.	Name of the Student	Title of the Project	Problem Understanding & Objectives (10 M)	Literature Survey (5 M)	Feasibility &Specification (5 M)	Presentation (5)	Total (25)	Signature of Supervisor

Signature of the HoD
Name: Dr. Y V R Naga Pawan

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT STAGE - I INTERNAL EVALUATION SHEET (REVIEW-II)

Year: IV Semester: I Program : B.Tech Branch: CSE Academic Year : 2024 – 25

Date(s) of Presentation:

S. No.	H.T.No.	Name of the Student	Title of the Project	System Architecture (5 M)	Module Design (5 M)	Design Representation (10M)	Presentation (5)	Total (25)	Signature of Supervisor

Signature of the HoD
Name: Dr. Y V R Naga Pawan

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PROJECT STAGE - I INTERNAL EVALUATION SHEET (Department Review Committee)

Year: IV Semester: I Program : B.Tech Branch: CSE Academic Year : 2024 – 25

Date(s) of Presentation:

S. No.	H.T.No.	Name of the Student	Title of the Project	Problem Understanding & Objectives (15 M)	System Architecture & Design (20M)	Presentation (15M)	Project Report (15)	Viva-Voce (10)	Total (75)

Signature of the Supervisor
Name:

:: 1 ::

Signature of the HoD
Name: Dr. Y V R Naga Pawan

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT STAGE - I INTERNAL EVALUATION SHEET (Department Review Committee)

Year: IV Semester: I Program : B.Tech Branch: CSE Academic Year : 2024 – 25

Date(s) of Presentation:

S. No.	H.T.No.	Name of the Student	Title of the Project	Problem Understanding & Objectives (15 M)	System Architecture & Design (20M)	Presentation (15M)	Project Report (15)	Viva-Voce (10)	Total (75)

Signature of the Senior Faculty
Name:

:: 1 ::

Signature of the HoD
Name: Dr. Y V R Naga Pawan

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT STAGE - I INTERNAL EVALUATION SHEET (Department Review Committee)

Year: IV Semester: I Program: B.Tech Branch: CSE Academic Year: 2024 – 25

Date(s) of Presentation:

S. No.	H.T.No.	Name of the Student	Title of the Project	Problem Understanding & Objectives (15 M)	System Architecture & Design (20M)	Presentation (15M)	Project Report (15)	Viva-Voce (10)	Total (75)

Signature of the HoD
Name: Dr. Y V R Naga Pawan

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT STAGE - II INTERNAL EVALUATION SHEET (REVIEW-1)

Year: IV Semester: II Program : B.Tech Branch: CSE Academic Year : 2024 – 25

Date(s) of Presentation:

S. No.	H.T.No.	Name of the Student	Title of the Project	Implementation (10 M)	Testing (10 M)	Presentation (5 M)	Total (25)	Signature of Supervisor

Signature of the HoD
Name: Dr. Y V R Naga Pawan

:: 1 ::

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT STAGE - II INTERNAL EVALUATION SHEET (REVIEW-II)

Year: IV Semester: II Program : B.Tech Branch: CSE Academic Year : 2024 – 25

Date(s) of Presentation:

S. No.	H.T.No.	Name of the Student	Title of the Project	Results (10 M)	Conclusion & Future Scope (10 M)	Presentation (5 M)	Total (25)	Signature of Supervisor

Signature of the HoD
Name: Dr. Y V R Naga Pawan

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT STAGE - II EXTERNAL EVALUATION SHEET LIST

Year: IV Semester: II Program : B.Tech Branch: CSE Academic Year : 2024 – 25

Date(s) of Presentation:

S. No.	H.T.No.	Name of the Student	Title of the Project	Presentation (15 M)	Execution (15M)	Results (15)	Project Report (20)	Viva-Voce (10)	Total (75 M)

Signature of the External Examiner
Name:

:: 1 ::

Signature of the HoD
Name: Dr. Y V R Naga Pawan

3.6 Evidence of Addressing Sustainable Development Goals (SDG) (10)

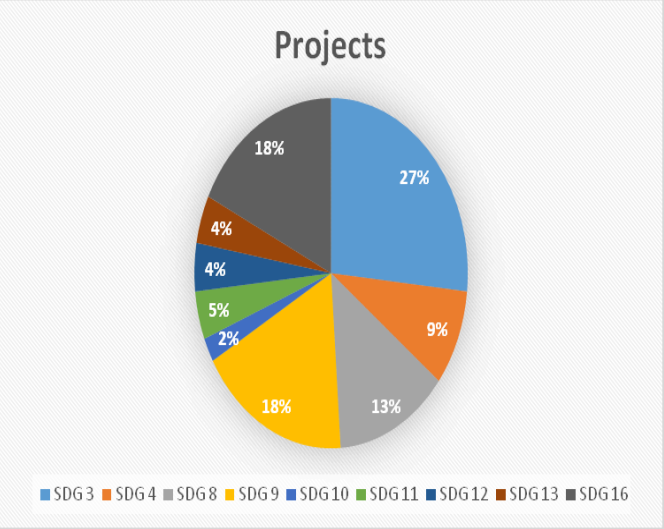
Total Marks 10.00

Institute Marks : 10.00

The Department of Computer Science and Engineering actively integrates the **United Nations Sustainable Development Goals (SDGs)** into academic activities, student projects, research initiatives, and community outreach programs. Through curriculum implementation, project-based learning, and technical activities, students are encouraged to develop innovative computing solutions that contribute to sustainable development and societal well-being.

1. SDG Integration through Student Projects

Students undertake **mini projects and major projects** addressing real-world problems aligned with SDGs.

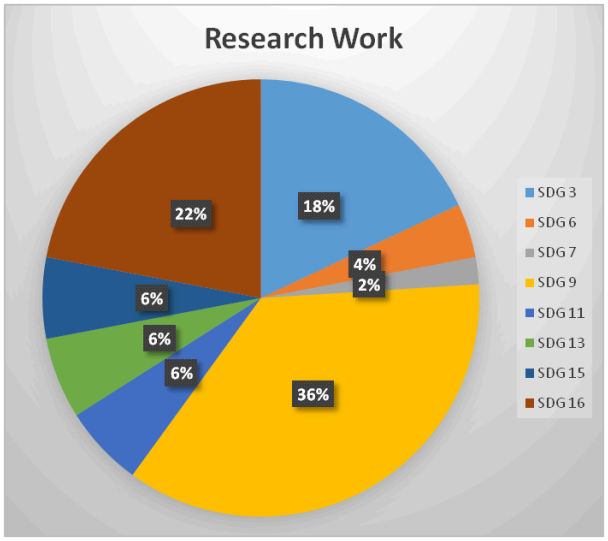


SDG	NUMBER OF PROJECTS
SDG 1	0
SDG 2	3
SDG 3	12
SDG 4	4
SDG 5	0
SDG 6	0
SDG 7	0
SDG 8	6
SDG 9	8
SDG 10	1
SDG 11	2
SDG 12	2
SDG 13	2
SDG 14	0
SDG 15	0
SDG 16	8
SDG 17	0

-

2. Research and Innovation Activities

Students and faculty are encouraged to carry out research activities in emerging areas such as Artificial Intelligence, Machine Learning, Internet of Things (IoT), Data Analytics, and Cloud Computing, which contribute to sustainable technological development. These research activities help in designing intelligent systems that improve efficiency, reduce resource consumption, and promote sustainable practices.



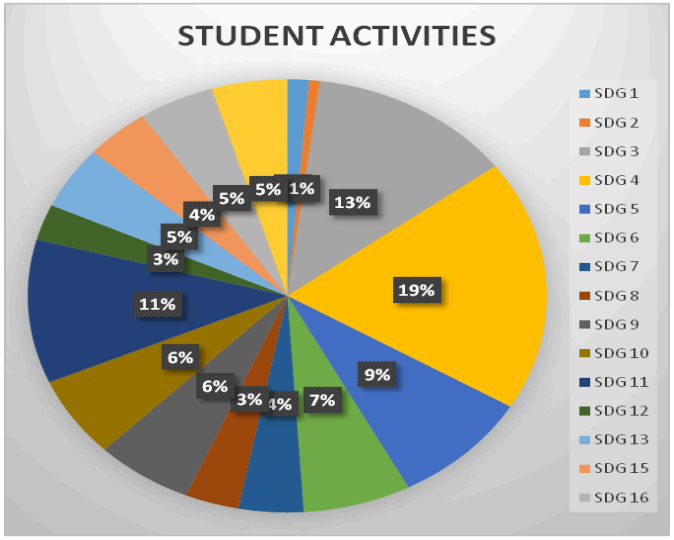
SDG	PUBLICATIONS
SDG 1	0
SDG 2	0
SDG 3	9
SDG 4	0
SDG 5	0
SDG 6	2
SDG 7	1
SDG 8	0
SDG 9	18
SDG 10	0
SDG 11	3
SDG 12	0
SDG 13	3
SDG 14	0
SDG 15	3
SDG 16	11
SDG 17	0

3. SDG-Oriented Workshops and Technical Events

The department organises seminars, workshops, hackathons, coding competitions, and technical talks focusing on themes such as Green Computing, Smart Cities, Digital Healthcare, and Sustainable Technologies. These activities motivate students to explore innovative solutions aligned with SDGs.

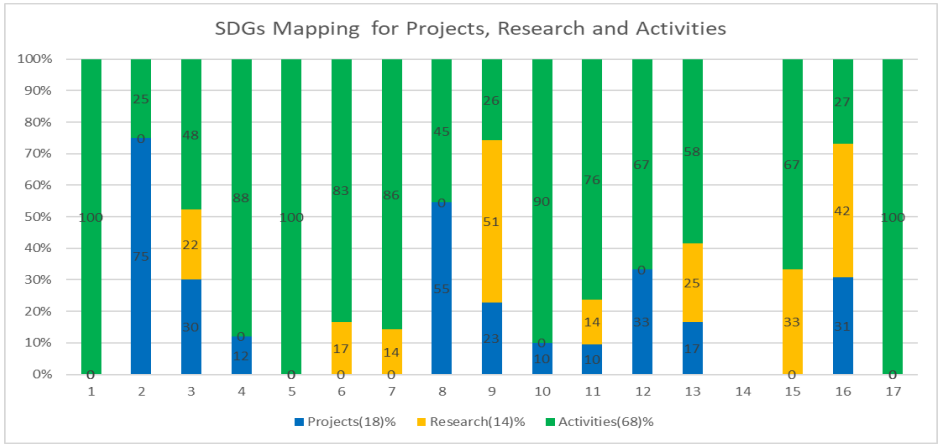
4. Community Engagement and Social Impact

Students participate in digital literacy programs, awareness campaigns, and community service activities aimed at improving technology accessibility and promoting sustainable practices among local communities. These initiatives support SDG 4 – Quality Education and encourage responsible use of technology for societal benefit.



SDG	STUDENT ACTIVITIES
SDG 1	2
SDG 2	1
SDG 3	19
SDG 4	28
SDG 5	13
SDG 6	10
SDG 7	6
SDG 8	5
SDG 9	9
SDG 10	9
SDG 11	16
SDG 12	4
SDG 13	7
SDG 14	0
SDG 15	6
SDG 16	7
SDG 17	7

TOTAL SDGS MAPPING:



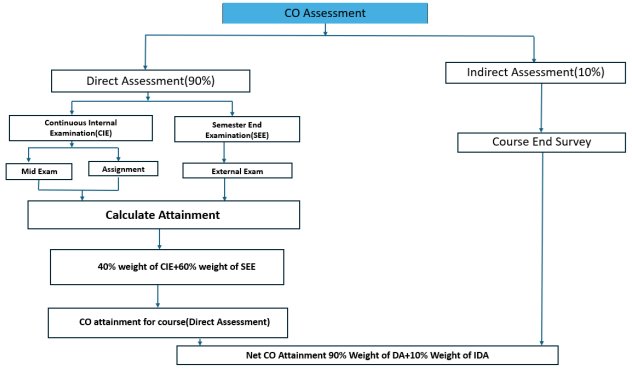
3.7 Attainment of Course Outcomes (25) — **Total Marks 25.00**

3.7.1. Describe the Assessment Tools and Processes Used to Gather the Data for the Evaluation of Course Outcome (5) — Institute Marks : 5.00

The assessment of Course Outcomes (COs) is a systematic process followed under Outcome-Based Education (OBE) to measure the extent to which students achieve the intended learning objectives of a course. As per the guidelines of the National Board of Accreditation, both direct and indirect assessment tools are used to gather data for CO evaluation.

- Direct assessment tools include Internal Assessment Tests, Semester End Examinations, assignments, quizzes, laboratory experiments, projects, and seminars. In these methods, each question or activity is mapped to specific COs and Bloom's Taxonomy levels. Student performance in CO-mapped questions is recorded and analysed to calculate attainment levels. Weightages are assigned to different components, such as Internal Assessment and Semester End Examination, to compute overall direct attainment.
- Indirect assessment tools include course-end surveys measure students' perception of learning and the relevance of the course outcomes. Survey responses are quantified and converted into attainment percentages. The overall CO attainment is calculated by combining direct and indirect assessment results using predefined weightages. If the target attainment level is not achieved, corrective actions such as remedial classes, revision of teaching strategies, improvement in assessment methods, and curriculum updates are implemented. Proper documentation, including CO-PO mapping, question paper analysis, attainment sheets, and action taken reports, is maintained to ensure transparency and continuous quality improvement in alignment with accreditation requirements

In Outcome Based Education, assessment is carried out by the department to identify, collect, analyze and evaluate the data towards the achievement of Course Outcomes (CO). The course outcomes are assessed based on direct assessment tools. The direct method of assessment includes the performance of the students in all the relevant assessment tools – Continuous Internal Examination (CIE) like Internal Assessments (IA), Assignments, Quizzes, and Semester End Examination (SEE). CIE contributes 20% and SEE contributes 80% to the total attainment of a course outcome



1. Direct Assessment Tools

Direct tools measure students' actual performance in examinations, assignments, and practical activities.

1.1 Semester End Examination (SEE)

- Conducted at the end of the semester.
- Includes descriptive, analytical, and problem-solving questions.
- Questions are mapped to specific COs and Bloom's Taxonomy levels.
- Marks obtained are used to calculate CO attainment based on predefined target levels.

1.2 Mid-Semester Tests (Internal Exams)

- Usually two mid examinations per semester.
- Assess conceptual understanding and problem-solving skills.

- The average of mid exams contributes to Continuous Internal Evaluation (CIE).
- Questions are CO-mapped and attainment is calculated separately for each CO.

1.3 . Rubrics for Laboratory and Project Assessments.

Rubrics are formulated for the assessment of Laboratory, Mini Project, Major projects, Seminar, and Internship courses. The attainment of Course Outcomes of all courses with respect to set attainment levels is recorded.

2. Indirect Assessment Tools

Indirect tools capture students' perceptions of learning and course effectiveness.

2.1 Course End Survey

- Students provide feedback on achievement of COs.
- Ratings are collected on a predefined scale (e.g., 1–5).
- Used to validate direct attainment results.

Direct Internal Co Attainment:

Course Outcomes	CO1	CO1	CO2	CO2	CO3	CO3	CO 1,2,3	CO3	CO3	CO4	CO4	CO5	CO5	CO 3,4,5	CO1	CO2	CO3	CO4	CO5	CO 1,2,3,4,5
Total number of students appeared for the examination (NST)	178	178	178	178	178	178	178	178	178	178	178	178	178	178	178	178	178	178	178	178

Course Outcomes	CO1	CO1	CO2	CO2	CO3	CO3	CO 1,2,3	CO3	CO3	CO4	CO4	CO5	CO5	CO 3,4,5	CO1	CO2	CO3	CO4	CO5	CO 1,2,3,4,5
Total number of students attempted the question (NSA)	148	109	147	58	148	100	177	100	77	155	50	151	97	174	173	173	173	172	172	174
Attempt % (TAP) = (NSA/NST)*100	83.15	61.24	82.58	32.58	83.15	56.18	99.44	56.18	43.26	87.08	28.09	84.83	54.49	97.75	97.19	97.19	97.19	96.63	96.63	97.75
Total number of Students who got more than 60% marks (NSM)	135	95	129	49	130	87	164	63	44	113	10	125	75	130	172	172	172	170	167	174
Attainment % (TMP) = (NSM/NSA)*100	91.22	87.16	87.76	84.48	87.84	87.00	92.66	63.00	57.14	72.90	20.00	82.78	77.32	74.71	99.42	99.42	99.42	98.84	97.09	100.00
Score(S)	3	3	3	3	3	3	3	2	2	3	0	3	3	3	3	3	3	3	3	3

Direct External Co Attainment:

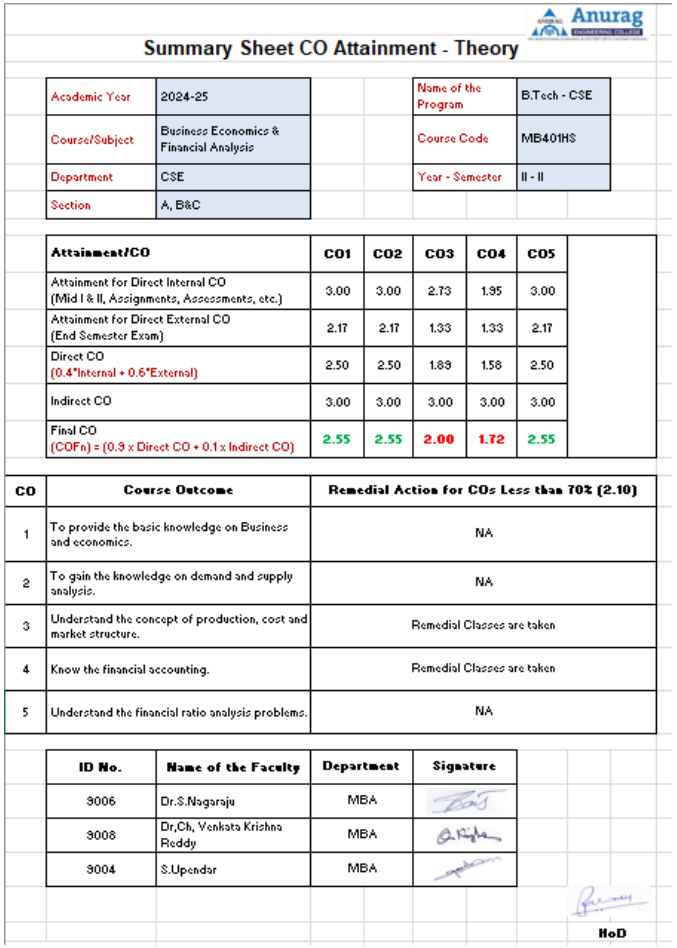
Course Outcomes	CO1	CO1	CO2	CO2	CO3	CO3	CO4	CO4	CO5	CO5	CO1	CO2	CO3	CO4	CO5
Total number of students appeared for the examination (NST)	171	171	171	171	171	171	171	171	171	171	171	171	171	171	171
Total number of students attempted the question (NSA)	157	153	155	131	142	153	125	136	158	81	171	171	151	171	171
Attempt % (TAP) = (NSA/NST)*100	91.81	89.47	90.64	76.61	83.04	89.47	73.10	79.53	92.40	47.37	100.00	100.00	88.30	100.00	100.00
Total number of Students who got more than 60% marks (NSM)	157	153	155	131	142	153	125	136	158	81	119	94	60	68	86
Attainment % (TMP) = (NSM/NSA)*100	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	69.59	54.97	39.74	39.77	50.29
Score(S)	3	3	3	3	3	3	3	3	3	3	2	2	1	1	2

Indirect Co Attainment:

Course Outcomes	CO1	CO2	CO3	CO4	CO5
Total number of students	136	136	136	136	136

appeared for the survey (NST)					
Total number of students attempted the question (NSA)	**136**	**136**	**136**	**136**	**136**
Attempt % (TAP) = (NSA/NST)*100	**100**	**100**	**100**	**100**	**100**
Total number of Students who gave more than 60% marks (NSM)	**125**	**128**	**129**	**130**	**129**
Attainment % (TMP) = (NSM/NSA)*100	**91.91**	**94.118**	**94.853**	**95.588**	**94.853**
Score(S)	**3**	**3**	**3**	**3**	**3**

Sample Attainment Summery Sheet



Summary Sheet CO Attainment - Theory

Academic Year	2024-25	Name of the Program	B.Tech - CSE
Course/Subject	Business Economics & Financial Analysis	Course Code	MB401HS
Department	CSE	Year - Semester	II - II
Section	A, B&C		

Attainment/CO	CO1	CO2	CO3	CO4	CO5
Attainment for Direct Internal CO (Mid I & II, Assignments, Assessments, etc.)	3.00	3.00	2.73	1.95	3.00
Attainment for Direct External CO (End Semester Exam)	2.17	2.17	1.33	1.33	2.17
Direct CO (0.4*Internal + 0.6*External)	2.50	2.50	1.89	1.58	2.50
Indirect CO	3.00	3.00	3.00	3.00	3.00
Final CO (COFn) = (0.9 x Direct CO + 0.1 x Indirect CO)	2.55	2.55	2.00	1.72	2.55

CO	Course Outcome	Remedial Action for COs Less than 70% (2.10)
1	To provide the basic knowledge on Business and economics.	NA
2	To gain the knowledge on demand and supply analysis.	NA
3	Understand the concept of production, cost and market structure.	Remedial Classes are taken
4	Know the financial accounting.	Remedial Classes are taken
5	Understand the financial ratio analysis problems.	NA

ID No.	Name of the Faculty	Department	Signature
9006	Dr.S.Nagaraju	MBA	
9008	Dr,Ch, Venkata Krishna Reddy	MBA	
9004	S.Upendar	MBA	

HoD

3.7.2 Record the Attainment of Course Outcomes of all Courses with Respect to Set Attainment Levels (20) | Institute Marks : 20.00

As part of the academic quality assurance process, Course Outcome (CO) attainment levels for each course are established at the beginning of the academic semester based on the defined course outcomes and their curriculum mapping. These levels are determined by considering the expected level of learning and competencies that students should achieve after the course is delivered by the concerned faculty.

The attainment of COs is assessed using both Continuous Internal Evaluation (CIE) and the Semester End Examination (SEE). The CIE consists of multiple assessment components such as mid-term examinations, assignments, quizzes, laboratory work, and project-based evaluations, each of which is carefully mapped to the relevant COs. The SEE evaluates the students' overall understanding and mastery of the course content and is also aligned with the respective COs.

Target attainment for each course is fixed to ensure that students achieve the desired level of learning outcomes. A standard target 70% is set as per institutional policy to maintain uniformity across all courses.The target is generally set at 70% attainment, meaning that at least 70% of students should score above the defined performance threshold in assessments mapped to Course Outcomes (COs). If the target attainment is not achieved, first analyze the performance to identify weak Course Outcomes and problem areas. Determine the root causes such as teaching gaps, difficult questions, or low student understanding. Provide remedial classes, extra practice, and improved study materials for students. Modify teaching methods and assessment strategies to better align with Course Outcomes.

Attainment of Course Outcomes for all Courses:

Year	Semester	Course Name	CO-1	CO-2	CO-3	CO-4	CO-5
I	I	Matrices and Calculus	2.36	2.36	2.94	3.00	2.10
I	I	Engineering Chemistry	2.81	2.71	2.85	3.00	2.90
I	I	Programming for Problem Solving	2.40	2.32	2.37	2.26	2.29
I	I	Basic Electrical Engineering	2.27	2.55	2.10	2.10	3.00
I	I	Computer Aided Engineering Graphics	2.66	2.66	2.87	3.00	3.00
I	I	Elements of Computer Science &Engineering	2.72	2.72	2.86	3.00	3.00
I	I	Engineering Chemistry Laboratory	3.00	3.00	3.00	3.00	3.00
I	I	Programming for Problem Solving Laboratory	2.82	2.82	2.82	2.82	2.82
I	I	Basic Electrical Engineering Laboratory	3.00	3.00	3.00	3.00	3.00
I	II	Ordinary Differential Equations and Vector Calculus	3.00	2.46	1.59	2.46	1.37
I	II	Applied Physics	2.98	2.53	2.99	2.98	2.97
I	II	Engineering Workshop	3.00	3.00	3.00	3.00	3.00
I	II	English for Skill Enhancement	2.77	2.54	2.69	2.61	2.61
I	II	Electronic Devices and Circuits	2.45	1.79	2.39	2.40	3.00
I	II	Applied Physics Laboratory	2.82	2.82	2.82	2.82	2.82
I	II	Python Programming Laboratory	1.58	1.58	1.54	1.48	1.48
I	II	English Language and Communication Skills Laboratory	3.00	3.00	3.00	3.00	3.00
I	II	IT Workshop	2.78	2.78	2.72	2.68	2.68

II	I	Probability and Statistics	3.00	2.46	3.00	2.55	2.10
II	I	Digital Electronics	2.40	2.43	2.61	2.57	2.59
II	I	Data Structures	2.61	2.33	3.00	3.00	3.00
II	I	Computer Organization and Architecture	2.43	2.14	2.47	2.23	2.23
II	I	Object Oriented Programming through Java	2.18	2.10	2.49	2.46	2.18
II	I	Data Structures Laboratory	2.81	2.81	2.86	2.91	2.91
II	I	Object Oriented Programming through Java Laboratory	2.68	2.68	2.72	2.78	2.78
II	II	Business Economics & Financial Analysis	2.55	2.55	2.00	1.72	2.55
II	II	Discrete Mathematics	2.91	2.46	2.96	2.39	2.94
II	II	Operating Systems	2.77	2.18	2.30	2.75	2.85
II	II	Database Management Systems	2.27	2.10	2.51	2.30	2.39
II	II	Software Engineering	2.10	2.10	2.47	2.49	2.34
II	II	Operating Systems Laboratory	2.90	2.90	2.89	2.90	2.90
II	II	Database Management Systems Laboratory	2.91	2.91	2.91	2.91	2.91
II	II	Real-time Research Project/ Societal Related Project	3.00	3.00	3.00	3.00	3.00
III	I	Design and Analysis of Algorithms	3.00	2.75	2.85	2.92	2.87
III	I	Computer Networks	2.81	2.46	2.95	2.10	2.10
III	I	DevOps	3.00	2.92	3.00	3.00	3.00
III	I	Scripting Languages (PE)	2.31	2.38	2.85	2.46	2.38

III	I	Principles of Programming Languages (PE)	2.26	2.43	2.52	2.61	2.61
III	I	Information Retrieval Systems (PE)	3.00	3.00	2.95	2.55	3.00
III	I	Distributed Databases (PE)	2.92	2.85	2.61	2.83	2.85
III	I	Computer Networks Laboratory	2.70	2.70	2.84	3.00	3.00
III	I	DevOps Laboratory	2.46	2.46	2.46	2.46	2.46
III	I	Advanced English Communication Skills Laboratory	3.00	3.00	3.00	3.00	3.00
III	II	Machine Learning	3.00	3.00	3.00	2.55	2.55
III	II	Formal Languages and Automata Theory	2.36	2.36	2.41	2.55	2.42
III	II	Artificial Intelligence	2.69	2.15	2.61	2.15	2.61
III	II	Internet of Things (PE)	3.00	3.00	2.95	2.91	3.00
III	II	Data Analytics (PE)	3.00	3.00	2.61	2.61	2.61
III	II	Disaster Preparedness & Planning Management (OE)	2.55	2.10	2.55	2.55	2.55
III	II	Renewable Energy Sources (OE)	3.00	3.00	3.00	3.00	3.00
III	II	Machine Learning Laboratory	3.00	3.00	3.00	3.00	3.00
III	II	Artificial Intelligence Laboratory	2.82	2.82	2.82	2.82	2.82
III	II	Internet of Things Laboratory (PE)	2.82	2.82	2.77	2.72	2.72
III	II	Data Analytics Laboratory (PE)	2.36	2.36	2.41	2.46	2.46
III	II	Industrial Oriented Mini Project/ Internship/Skill Development Course (Big data-Spark)	3.00	3.00	3.00	3.00	3.00
IV	I	Management Science	2.94	2.52	2.12	2.52	2.52

IV	I	Linux Programming	2.30	2.69	2.45	2.58	2.79
IV	I	Multimedia Computing (PE)	3.00	3.00	3.00	3.00	3.00
IV	I	Big Data Analytics (PE)	3.00	3.00	3.00	3.00	3.00
IV	I	Cloud Computing (PE)	2.92	2.84	2.93	3.00	3.00
IV	I	Artificial Intelligence (PE)	3.00	3.00	3.00	3.00	3.00
IV	I	Disaster Management (OE)	2.58	3.00	3.00	3.00	3.00
IV	I	Linux Programming Lab	3.00	3.00	3.00	3.00	3.00
IV	I	Mini project Evaluation	3.00	3.00	3.00	3.00	3.00
IV	I	Project Stage-I	3.00	3.00	3.00	3.00	3.00
IV	II	Internet of Things (PE)	2.45	2.54	2.41	2.48	2.03
IV	II	Software Project Management (PE)	2.58	2.51	2.29	2.87	3.00
IV	II	Mobile Computing (PE)	2.58	2.50	2.58	2.58	2.58
IV	II	Cyber Forensics (PE)	2.16	2.16	2.16	2.16	2.16
IV	II	Project Management (OE)	3.00	3.00	2.58	3.00	2.58
IV	II	Safety Engineering (OE)	3.00	2.58	3.00	3.00	3.00
IV	II	Seminar	3.00	3.00	3.00	3.00	3.00
IV	II	Project Stage-II	3.00	3.00	3.00	3.00	3.00

3.8 Attainment of Program Outcomes and Program Specific Outcomes (25) — **Total Marks 25.00**

Institute Marks : 25.00

PO Attainment

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
MA101BS	2.50	2.55	2.50	2.50	2.50	PO6	PO7	PO8	PO9	PO10	2.50	PO12
CH102BS	2.90	2.90	2.91	2.83	2.91	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS103ES	2.32	2.31	2.32	PO4	2.32	PO6	PO7	PO8	PO9	PO10	PO11	PO12
EE104ES	2.35	2.31	2.38	2.28	2.30	2.43	PO7	PO8	PO9	PO10	PO11	PO12
EG105ES	2.82	2.62	2.62	2.95	2.62	PO6	PO7	PO8	PO9	2.62	PO11	PO12
CS106ES	2.84	2.86	2.86	2.88	2.88	3.00	3.00	PO8	PO9	PO10	2.88	PO12
CH107BS	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	PO12
CS109ES	2.82	2.82	2.82	2.82	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
MA201BS	2.08	2.04	1.93	2.16	2.08	PO6	PO7	PO8	PO9	PO10	2.08	PO12
EE108ES	3.00	3.00	3.00	3.00	3.00	PO6	PO7	PO8	PO9	PO10	3.00	PO12
AP202BS	2.87	2.76	2.48	2.98	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
ME203ES	3.00	3.00	PO3	3.00	3.00	PO6	PO7	PO8	3.00	PO10	3.00	PO12
EN204HS	PO1	PO2	PO3	PO4	PO5	PO6	2.60	2.57	2.58	2.62	2.59	PO12
EC205ES	2.41	2.54	2.45	PO4	PO5	2.48	1.81	PO8	PO9	PO10	PO11	PO12
AP206BS	2.67	2.80	PO3	PO4	PO5	PO6	PO7	PO8	2.80	2.80	PO11	PO12
CS207ES	1.36	1.36	1.36	1.36	1.31	PO6	PO7	PO8	PO9	PO10	PO11	PO12
EN208HS	PO1	PO2	PO3	PO4	PO5	PO6	3.00	3.00	3.00	3.00	3.00	PO12
CS209ES	2.71	2.71	2.71	2.69	2.70	2.69	2.69	2.68	2.64	2.64	2.64	PO12
PS301BS	2.62	2.61	2.47	2.53	2.53	2.50	PO7	PO8	PO9	PO10	2.53	PO12
DE302ES	2.45	2.48	2.47	2.47	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS303PC	2.77	2.78	2.82	3.00	2.83	PO6	PO7	2.63	PO9	PO10	2.87	PO12
CS304PC	2.23	2.20	2.18	2.11	2.17	2.14	PO7	PO8	PO9	2.09	2.21	PO12
CS305PC	2.18	2.21	2.21	2.44	2.23	2.09	2.09	2.24	2.24	2.09	2.18	PO12
CS306PC	2.84	2.85	2.86	PO4	PO5	PO6	PO7	2.87	PO9	PO10	2.85	PO12
CS307PC	2.69	2.69	2.70	2.69	2.70	2.75	2.75	2.71	2.70	2.75	2.69	PO12
MB401HS	2.20	2.22	2.20	2.01	2.04	2.26	2.25	2.50	2.25	2.12	2.20	PO12
CS402PC	2.70	2.69	2.72	2.72	2.68	PO6	PO7	PO8	PO9	PO10	2.70	PO12
CS403PC	2.53	2.47	2.65	2.15	2.54	PO6	PO7	2.74	2.22	PO10	2.62	PO12
CS404PC	2.24	2.22	2.26	2.20	2.30	PO6	2.22	PO8	2.46	PO10	2.24	PO12

CS405PC	2.22	2.24	2.25	2.28	2.22	2.18	2.17	2.27	PO9	PO10	2.25	PO12
CS406PC	2.89	2.89	2.89	2.89	2.89	PO6	PO7	2.89	2.89	PO10	PO11	PO12
CS407PC	2.90	2.90	2.90	2.90	2.90	PO6	PO7	2.90	2.90	PO10	2.90	PO12
CS408PW	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	PO12
CS501PC	2.87	2.86	2.87	2.85	2.87	PO6	PO7	PO8	PO9	PO10	2.87	PO12
CS502PC	2.46	2.43	2.67	2.89	2.42	PO6	PO7	PO8	2.00	PO10	2.58	PO12
CS503PC	2.98	2.98	2.98	3.00	2.99	3.00	3.00	3.00	2.98	3.00	2.99	PO12
CS513PE	2.29	2.53	2.31	2.31	2.26	2.31	2.31	2.83	2.42	2.40	2.31	PO12
CS515PE	2.43	2.39	2.44	2.46	2.43	2.43	2.43	2.42	2.41	2.43	2.43	PO12
CS523PE	2.88	2.87	2.97	2.96	2.91	PO6	PO7	PO8	2.55	PO10	2.98	PO12
CS524PE	2.79	2.78	2.78	2.77	2.77	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS504PC	2.79	2.82	2.85	2.81	2.91	2.83	2.91	2.83	2.82	2.83	2.83	PO12
CS505PC	2.40	2.40	2.40	2.40	2.40	PO6	PO7	2.40	2.40	2.40	PO11	PO12
AE506HS	PO1	PO2	PO3	PO4	PO5	PO6	3.00	3.00	3.00	3.00	3.00	PO12
CS601PC	2.80	2.79	2.79	2.77	2.75	3.00	PO7	PO8	PO9	PO10	2.75	PO12
CS603PC	2.38	2.35	2.28	2.35	2.33	PO6	PO7	2.06	PO9	PO10	2.40	PO12
CS632PE	2.97	2.96	2.98	2.97	2.97	2.95	PO7	PO8	3.00	3.00	2.96	PO12
CS633PE	2.77	2.75	2.70	2.76	2.73	2.57	PO7	PO8	PO9	2.57	PO11	PO12
CE611OE	2.36	2.00	2.50	2.00	2.50	2.50	2.31	2.25	2.50	2.30	2.50	PO12
EE611OE	3.00	3.00	3.00	3.00	3.00	3.00	3.00	PO8	PO9	3.00	3.00	PO12
CS604PC	3.00	3.00	3.00	3.00	3.00	3.00	PO7	3.00	3.00	3.00	3.00	PO12
CS605PC	2.80	2.80	2.80	2.80	2.80	2.80	2.80	2.80	2.80	2.80	2.80	PO12
CS642PE	2.74	2.74	2.74	PO4	2.74	2.80	PO7	PO8	2.74	2.69	2.69	PO12
CS606PW	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	PO12
CS702PC	2.77	2.78	2.78	2.84	2.78	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS731PE	3.00	3.00	3.00	3.00	3.00	3.00	PO7	PO8	3.00	PO10	3.00	PO12
CS732PE	3.00	3.00	3.00	3.00	3.00	3.00	PO7	PO8	PO9	3.00	PO11	PO12
CS742PE	2.71	2.80	2.74	2.76	3.00	2.71	3.00	2.74	2.77	3.00	2.41	PO12
CS743PE	3.00	3.00	3.00	3.00	3.00	3.00	PO7	3.00	3.00	3.00	3.00	PO12
CE721OE	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	PO12
CS704PW	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	PO12
CS852PE	2.78	2.74	2.68	2.72	2.83	PO6	PO7	PO8	PO9	2.84	PO11	PO12

CS853PE	2.57	2.52	2.47	2.66	2.22	PO6	PO7	2.93	2.81	2.84	2.59	PO12
CS861PE	2.50	2.52	2.51	2.51	2.53	2.50	2.53	2.53	2.52	2.53	2.45	PO12
CS862PE	2.07	2.07	2.07	2.07	2.07	2.07	2.07	2.07	2.07	2.07	2.07	PO12
CE831OE	2.84	2.87	2.77	2.84	2.83	2.69	3.00	2.77	2.71	2.81	2.93	PO12
CE832OE	3.00	3.00	2.83	2.87	2.77	2.84	3.00	3.00	2.88	2.91	3.00	PO12
CS801PW	3.00	3.00	3.00	3.00	3.00	3.00	PO7	3.00	3.00	3.00	3.00	PO12
MS701HS	2.77	2.80	2.77	2.77	2.77	2.77	2.77	2.83	2.78	2.77	2.77	PO12
CS703PC	3.00	3.00	3.00	3.00	3.00	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CS602PC	2.35	2.36	2.37	2.36	2.36	PO6	PO7	2.36	PO9	PO10	PO11	PO12
CS643PE	2.34	2.34	2.35	2.34	2.35	2.40	PO7	2.40	2.40	2.40	PO11	PO12
CS802PW	PO1	PO2	PO3	PO4	PO5	3.00	3.00	3.00	3.00	3.00	3.00	PO12
CS705PW	PO1	PO2	PO3	PO4	PO5	3.00	3.00	3.00	3.00	3.00	3.00	PO12

PO Attainment Indirect

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PROGRAM EXI	2.32	2.46	2.46	2.53	2.45	2.52	2.46	2.46	2.49	2.49	2.55	PO12

PO Attainment Level

> Note: The Institution can fix the weightage of the indirect attainment maximum up to 20%.

Define the Weightage for Indirect Attainment: 0.20

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
Direct Attainment	2.68	2.68	2.67	2.69	2.67	2.72	2.70	2.73	2.73	2.74	2.72	0
InDirect Attainment	2.32	2.46	2.46	2.53	2.45	2.52	2.46	2.46	2.49	2.49	2.55	0
Overall Attainment	2.68	2.68	2.67	2.69	2.67	2.72	2.70	2.73	2.73	2.74	2.72	0

PSO Attainment

Course	PSO1	PSO2
MA101BS	2.51	2.50
CH102BS	2.79	2.75
CS103ES	2.32	2.32
EE104ES	2.59	2.43
EG105ES	2.86	2.75

CS106ES	2.86	2.88
CH107BS	3.00	3.00
CS109ES	2.82	2.82
EE108ES	3.00	3.00
MA201BS	2.28	2.08
AP202BS	2.09	PSO2
ME203ES	PSO1	3.00
EC205ES	1.65	PSO2
AP206BS	PSO1	2.80
CS207ES	1.36	1.35
EN208HS	PSO1	3.00
CS209ES	2.72	2.69
EN204HS	PSO1	2.60
PS301BS	2.60	2.55
DE302ES	2.47	2.47
CS305PC	2.21	2.21
CS306PC	2.85	2.85
CS307PC	2.69	2.70
MB401HS	2.14	2.20
CS402PC	2.70	2.72
CS404PC	2.24	2.26
CS405PC	2.24	2.22
CS406PC	2.89	2.89
CS407PC	2.90	2.90
CS408PW	3.00	3.00
CS503PC	2.98	2.98
CS515PE	2.45	2.40
CS524PE	2.79	2.78
CS505PC	2.40	2.40
AE506HS	PSO1	3.00
CS601PC	2.81	2.73
CS633PE	2.77	2.75
EE611OE	3.00	3.00
CS604PC	3.00	3.00

CS605PC	2.80	2.80
CS606PW	3.00	3.00
MA701HS	2.78	2.77
CS731PE	3.00	3.00
CS742PE	2.81	2.71
CS743PE	3.00	3.00
CE721OE	3.00	3.00
CS703PC	3.00	3.00
CS704PW	3.00	3.00
CS705PW	3.00	3.00
CS852PE	2.75	2.73
CS853PE	2.58	2.58
CS861PE	2.52	2.52
CS862PE	2.07	2.07
CS801PW	3.00	3.00
CS802PW	3.00	3.00
CS602PC	2.35	2.37
CS643PE	2.34	2.35
CS732PE	3.00	3.00
CS303PC	2.81	2.82
CS304PC	2.23	2.24
CS403PC	2.54	2.56
CS501PC	2.86	2.87
CS502PC	2.36	2.39
CS513PE	2.44	2.41
CS523PE	2.87	2.91
CS504PC	2.90	2.89
CS603PC	2.36	2.39
CS632PE	2.96	2.96
CE611OE	2.39	2.40
CS642PE	2.73	2.74
CS702PC	2.81	2.80
CE831OE	2.74	2.77
CE832OE	2.92	2.91

PSO Attainment Indirect

Survey	PSO1	PSO2
PROGRAM EXIT SURVEY	2.44	2.51

PSO Attainment Level

Course	PSO1	PSO2
Direct Attainment	2.66	2.69
InDirect Attainment	2.44	2.51
Overall Attainment	2.66	2.69